

**Name of the challenge:** Vibe-code a real physics simulation app

**One-line brief:**

Use Quanscient's Allsolve, a cloud multiphysics simulation engine, to create a tool that runs a real physics simulation for a use case you identify.

**Why this challenge matters:**

Engineering simulation usually means one expert working with complex desktop software. Quanscient Allsolve SDK exposes that same simulation as code, so anyone can build a custom app for a specific problem. AI coding tools make this a few hours' work instead of months. We want to see what our SDK can enable: new use cases, interfaces, and ways to solve and optimize real engineering problems faster.

**What a great solution looks like:**

A working app (rough is fine) that takes user input, calls Allsolve via the SDK, runs a real simulation and returns a result the user can act on. The physics underneath must be genuinely doing the work (not a fake number dressed up in a nice UI). We'll provide Allsolve access and web app examples.

The solution should address a realistic and meaningful use case. Show the real industry problem it solves and how a company in that field could benefit from an Allsolve SDK that enables them to build apps like yours.

Your experience with the SDK is also part of the challenge. Critically evaluate its usability and ease of use, and identify concrete ideas for improvement.

**Winning criteria (ranked):**

1. A meaningful industry problem, which provides value for the user
2. Creativity / originality of the use case
3. SDK experience and insights; how clearly solution identifies SDK usability, pain points, and opportunities for improvement
4. A working demo, physics actually used and logic seems correct